

Homework (Carolina Reaper)

Q1. Given $f(x) = x^2$ and $g(x) = x + 2$, what is $f(g(x))$?

- (A) $x^2 + 4x + 4$
- (B) $x^2 + 2x + 4$
- (C) $x^2 - 4x + 4$
- (D) $x^2 + 4x - 4$
- (E) $x^2 - 2x - 4$

Q2. A music producer wants to create a rhythm pattern using a combination of two functions, $f(x)$ and $g(x)$, where $f(x)$ is the number of beats per minute and $g(x)$ is the duration of each beat in seconds. If $f(x) = 120$ and $g(x) = x + 1$, what is the composite function $f(g(x))$?

- (A) $120(x + 1)$
- (B) $120(x - 1)$
- (C) $120(x + 2)$
- (D) $120(x - 2)$
- (E) $120x$

Q3. The frequency of a sound wave is given by $f(x) = 20x$ Hz, where x is the amplitude of the wave. If the amplitude is adjusted by a factor of $g(x) = x - 2$, what is the domain of the composite function $f(g(x))$?

- (A) $x \geq 2$
- (B) $x \leq 2$
- (C) $x > 2$
- (D) $x < 2$
- (E) $x = 2$

Q4. An artist is creating a painting with a unique pattern of shapes. The size of each shape is determined by the composite function $f(g(x))$, where $f(x) = x^2$ and $g(x) = x + 3$. What is the size of the shape when $x = 2$?

- (A) 25
- (B) 16
- (C) 36
- (D) 9
- (E) 1

Q5. A musician is composing a piece of music using a com-

bination of two functions, $f(x)$ and $g(x)$, where $f(x)$ is the melody and $g(x)$ is the harmony. If $f(x) = 2x$ and $g(x) = x - 1$, what is the composite function $f(g(x))$?

- (A) $2x - 2$
- (B) $2x + 2$
- (C) $2x - 1$
- (D) $2x + 1$
- (E) $x - 1$

Q6. The volume of a sound wave is given by $f(x) = x^3$ cubic meters, where x is the frequency of the wave. If the frequency is adjusted by a factor of $g(x) = x + 4$, what is the domain of the composite function $f(g(x))$?

- (A) $x \geq -4$
- (B) $x \leq -4$
- (C) $x > -4$
- (D) $x < -4$
- (E) $x = -4$

Q7. An art gallery is displaying a series of paintings with unique patterns. The size of each pattern is determined by the composite function $f(g(x))$, where $f(x) = x^2 + 1$ and $g(x) = x - 2$. What is the size of the pattern when $x = 3$?

- (A) 16
- (B) 10
- (C) 12
- (D) 8
- (E) 4

Q8. A music teacher is creating a lesson plan using a combination of two functions, $f(x)$ and $g(x)$, where $f(x)$ is the tempo and $g(x)$ is the time signature. If $f(x) = 3x$ and $g(x) = x + 2$, what is the composite function $f(g(x))$?

- (A) $3x + 6$
- (B) $3x - 6$
- (C) $3x + 2$
- (D) $3x - 2$
- (E) $x + 2$

Open Ended Questions (FRQ)

Q9. A painter is creating a mural with a unique pattern of shapes. The size of each shape is determined by the composite function $f(g(x))$, where $f(x) = x^2 - 2x + 1$ and $g(x) = x + 1$. Find the size of the shape when $x = 4$ and explain your reasoning.

Q10. A musician is composing a piece of music using a combination of two functions, $f(x)$ and $g(x)$, where $f(x)$ is the melody and $g(x)$ is the harmony. If $f(x) = 2x - 1$ and $g(x) = x + 3$, find the composite function $f(g(x))$ and determine its domain. Explain your reasoning.

Chef's Tips

- Ensure students understand the concept of composite functions and their applications in real-world contexts.
- Encourage students to use graphing calculators or software to visualize the composite functions and explore their properties.
- Remind students to check their work carefully and explain their reasoning clearly.